

# ExpenseSense: Privacy-Preserving On-Device Personal Finance Tool Calling with Sub-2B LLMs

Anonymous ACL submission

## Abstract

Personal finance data - detailed records of when, where, and how a user spends money - is among the most sensitive information individuals possess, making cloud-based LLM inference undesirable for many users. Yet natural-language personal finance assistants currently require transmitting these queries to cloud-hosted LLMs, exposing users to data retention and third-party data flows. We introduce **ExpenseSense**, a personal finance Q&A assistant that runs entirely on-device, ensuring that no financial records, query text, or model outputs leave the user’s device. To evaluate whether this privacy architecture is practically viable, we benchmark on-device tool calling using a 115-question test suite across six sub-2B quantized SLMs (0.8B-2B), comparing single-agent (joint routing and parameter extraction) against dual-agent (router-specialist) architectures. Architectural decomposition substantially improves both accuracy and safety: dual-agent Qwen3.5-2B achieves 77.6% strict Correct Ratio (+17.4 pp over single-agent) while reducing average latency by 48%, and dual-agent Qwen3.5-0.8B reaches 89.1% Task Accuracy in 3.0 s with 1.2 GB RAM. Critically, dual-agent architectures shift errors from spurious parameters to safer, missing parameters, reducing the risk of silent incorrect outputs. Our results demonstrate that private, high-quality financial analytics is achievable on consumer hardware today, with no cloud dependency.

## 1 Introduction

Large language models (LLMs) have rapidly advanced tool-augmented AI (Schick et al., 2023; Patil et al., 2023; Qin et al., 2024), enabling agents to invoke structured APIs from natural language. However, research primarily targets large cloud-hosted models which is problematic for personal finance where local inference is necessary for privacy. Expense records reveal sensitive behavioral

patterns (e.g., health-related purchases). Users expect natural language interfaces (e.g., asking “how much did I spend on food last month?”) that map to structured queries over local data (Li et al., 2024). This creates an underexplored problem: can *sub-2B quantized models* running entirely on consumer hardware reliably call domain-specific financial tools from informal, colloquial language, and in doing so, serve as practical privacy infrastructure for real users?

Existing benchmarks (Qin et al., 2024; Zhu et al., 2026; Cao et al., 2026) focus on models above 7B parameters, which are prohibitively slow for local deployment. Conversely, on-device models like Octopus v2 (Chen and Li, 2024) and Hammer (Lin et al., 2024) demonstrate the potential of smaller models but have not been evaluated on domain-specific financial schemas or under consumer-hardware latency constraints.

We address three research questions (**RQ**):

- **RQ1:** Does on-device inference with sub-2B SLMs provide a practically viable privacy protection for personal finance tool use, i.e., can users obtain accurate financial analytics without surrendering query data to cloud providers?
- **RQ2:** Does decomposing intent classification from parameter extraction (dual-agent) yield significant accuracy gains over a joint single-agent architecture?
- **RQ3:** What are the primary failure modes across architectures, and do they differ in the human-facing harms they produce?

This paper introduces **ExpenseSense**, with the following contributions:

1. Empirical **on-device deployment guidelines for privacy**: Qwen3.5-0.8B in dual mode achieves 89.1% Task Accuracy with 3.0 s latency and  $\approx 1.2$  GB RAM, demonstrating that

- private, high-quality personal finance analytics can run locally on consumer hardware making on-device inference a practical privacy protection feature.
2. A **fine-grained error taxonomy** of 7 categories showing that dual-agent architectures consolidate errors into predictable, safer failure modes-shifting from spurious parameter additions (which silently corrupt query results and can mislead the user) toward missing parameters (which are visible omissions). This error-mode shift is the central human-safety argument for architectural decomposition.
  3. A comparative evaluation of **single-agent vs. dual-agent** architectures across six quantized SLMs (0.8B-2.0B parameters), showing that dual-agent decomposition reduces latency (by up to 48% for Qwen3.5-2B), improves exact-match correctness (+17.4 pp Correct Ratio), and stabilizes sub-1B models without cloud resources.
  4. A **115-question dataset** for financial tool calling across five analytical tools, with a 7-feature complexity scoring function categorizing queries into three difficulty tiers (L1-L3) that simulate real-world user interactions.

## 2 On-Device Inference as Privacy Protection for Real Users

The deployment of LLMs in personal finance exposes a structural tension: the same conversational interface that makes financial analytics accessible to users also makes their sensitive behavioral data accessible to cloud providers. When a user types "how much did I spend on therapy last month?" into a cloud-hosted assistant, they are not merely querying a database-they transmit evidence of a health condition, a temporal pattern, and financial commitment to a third-party infrastructure they did not choose and cannot audit. The query is a disclosure.

Staab et al. (2024) demonstrated that LLMs can infer sensitive personal attributes-including location, income, and demographic characteristics-from seemingly innocuous texts, and Yao et al. (2024) catalog privacy risks including data leakage, memorization, and inference attacks. The implication for personal financial assistants is direct: even if the database never leaves the device, routing natural-language queries through cloud LLMs

leaks behavioral signals that can be retained, aggregated, and exploited beyond the user's control or knowledge.

ExpenseSense addresses this threat by adopting an *on-device, privacy-first* architecture. By leveraging sub-2B quantized models that run efficiently on consumer hardware (tested on an Apple M1 with 8 GB unified memory), all user data, query text, and model outputs remain strictly local. No financial records, queries or inferences leave the device.

The local-first design preserves *contextual integrity* (Nissenbaum, 2004) by ensuring financial queries and data remain within their original context.

The practical threat model covers three risks: cloud data retention, query-level inference attacks (Staab et al., 2024), and third-party exposure. On-device inference eliminates all three by ensuring queries never leave for external infrastructure.

The key challenge is that on-device inference with sub-2B models has historically been unreliable for structured tool calling (Kavathekar et al., 2025). Our work demonstrates that architectural decomposition-separating intent classification from parameter extraction-bridges this reliability gap, making private on-device inference not merely possible but practical.

## 3 Related Work

**Tool-Augmented Language Models.** Toolformer (Schick et al., 2023) demonstrated self-supervised tool-use learning; Gorilla (Patil et al., 2023) and ToolLLM (Qin et al., 2024) scaled to thousands of real-world APIs. HuggingGPT (Shen et al., 2023) and ToolAlpaca (Tang et al., 2023) explored multi-step orchestration. APIGen (Liu et al., 2024b) introduced a verified pipeline for generating function-calling training data, showcasing that 1B-parameter models fine-tuned on high-quality synthetic data can surpass GPT-3.5-Turbo on structured tool invocation. These approaches rely on cloud-hosted models ( $\geq 7B$ ), leaving the privacy-constrained on-device setting largely unaddressed.

**Function-Calling Evaluation.** The Berkeley Function Calling Leaderboard (Patil et al., 2025) established a benchmark using AST substring matching and execution-based validation. Kavathekar et al. (2025) proposed an evaluation suite (FSP, ACS, AVC, Task Accuracy, and Correct Ratio) to evaluate small model function calling; we adopt

these metrics directly. CallNavi (Song et al., 2025) proposed a two-stage router+specialist approach similar to our dual-agent design-but evaluates models  $\geq 7B$  on general-purpose APIs. In contrast, we focus on sub-2B models for domain-specific, single-turn personal finance related tasks.

**On-Device Function Calling.** MobileLLM (Liu et al., 2024a), Phi-3 (Abdin et al., 2024), and related surveys (Xu et al., 2024) have explored sub 4B parameter models for mobile deployment. Octopus v2 (Chen and Li, 2024) fine-tuned a 2B model for on-device API calling via functional tokens, achieving competitive latency against GPT-4 on specific tasks. Hammer (Lin et al., 2024) introduced function masking to reduce overfitting to function names, addressing an issue similar to the category normalisation errors we observe. Apple Intelligence (Gunter et al., 2024) validated on-device tool calling at scale with a  $\sim 3B$  model, demonstrating the viability of the paradigm. Kavathekar et al. (2025) found that structured output compliance is the primary bottleneck for SLMs.

**Financial NLP and Privacy.** FinQA (Chen et al., 2021) and FinGPT (Yang et al., 2023) target financial numerical reasoning and generation, but focus on text generation rather than tool calling. FinMCP-Bench (Zhu et al., 2026) and FinTrace (Cao et al., 2026) evaluate LLM agents for financial tool use, but target cloud-scale models. Personal finance assistants (Li et al., 2024) require strict privacy making local execution crucial.

**Human-Centered Privacy for Language Models.** A growing body of work has seen privacy not as a technical property of models but as a human-centered concern shaped by user expectations and social context. Nissenbaum (2004)’s contextual integrity framework can be applied to conversational AI to show that data flows violate privacy when they cross contextual norms, even when no explicit PII is transmitted. Staab et al. (2024) demonstrated that LLMs act as inference engines, reconstructing sensitive personal attributes from ordinary text—a threat particularly acute when personal finance queries are routed through cloud providers. Yao et al. (2024) surveys the resulting landscape of data leakage, memorization, and inference risks in deployed LLM systems. Our work operationalises these concerns at the system level: rather than studying how cloud LLMs violate privacy, we build and evaluate an architecture that eliminates the ex-

| Tool                 | ID | Key Params                                                   |
|----------------------|----|--------------------------------------------------------------|
| plot_time_series     | 1  | category, year/month, ranges, months, ignore_rent            |
| plot_distribution    | 2  | category, year/month/day, ranges, months, ignore_rent        |
| plot_comparison_bars | 3  | category, y1/m1, y2/m2, sm/em/ey, ignore_rent                |
| calculate_total      | 4  | category, year/month/day, remarks, months, ignore_rent       |
| get_top_expenses     | 5  | n, category, year/month/day, months, min_amount, ignore_rent |

Table 1: Tool inventory. The five tools span visualisation (1–3), aggregation (4), and retrieval (5).

posure pathway entirely, keeping all query data local to the user’s device.

## 4 System Design

### 4.1 Tool Inventory

ExpenseSense exposes five analytical tools over a user’s expense database. Table 1 summarises each tool and its parameter space. These tools span core personal finance use cases: time series analysis, categorical breakdown, period comparison, aggregation, and ranked retrieval.

### 4.2 Architectural Configurations

We evaluate two local architectures for mapping natural language user queries to tool calls.

**Single-Agent Pipeline.** The single-agent pipeline uses a single prompt with all five tool definitions, parameter schemas, and joint few-shot examples. The model must select the tool and extract parameters in a single pass, outputting a JSON object containing the tool ID (1–5) and arguments for the chosen tool. The prompt includes structured output rules, date-handling instructions, and 73 tool-specific examples aggregated across all 5 tools. Prompt length:  $\approx 5,500$  tokens (context window inclusive).

**Dual-Agent Pipeline.** The task is decomposed into two sequential inference calls:

1. **Router:** A focused intent-classification prompt with 19 few-shot examples and explicit disambiguation rules (e.g., “compare” requires two distinct time periods) elicits a single digit (1–5).
2. **Specialist:** The predicted tool ID selects a tool-specific prompt containing 13–16 examples for that *one* tool, along with detailed parameter definitions and date-handling rules.

| Task Group      | Cases      | Avg Params | Avg $C$    |
|-----------------|------------|------------|------------|
| Time Series     | 23         | 2.9        | 7.0        |
| Distribution    | 21         | 2.5        | 6.4        |
| Comparison      | 28         | 4.4        | 8.8        |
| Calculate Total | 22         | 2.4        | 6.3        |
| Top Expenses    | 21         | 3.2        | 6.7        |
| <b>Total</b>    | <b>115</b> | <b>3.2</b> | <b>7.1</b> |

Table 2: Test case distribution by group. Comparison queries are the most parameter-dense and complex.

The router sees  $\approx 1,000$  tokens; the specialist sees  $\approx 1,500$ – $2,000$  tokens with deeper, more directly relevant supervision. This simplifies the task for each model call and provides more relevant in-context examples, though it requires two sequential inference passes. Combined prompt tokens:  $\approx 2,700$  per query.

### 4.3 Post-Hoc Parameter Validation

A post-processing validation layer corrects category names deterministically without additional model calls:

1. **Exact match** against the known vocabulary.
2. **Morphological normalisation**:  
groceries  $\rightarrow$  grocery, cafés  $\rightarrow$  cafe.
3. **Edit-distance matching** via difflib (cutoff  $\geq 0.80$ ).
4. **Token overlap scoring** (Jaccard; cutoff  $\geq 0.50$ ).

All results report both raw (pre-validation) and validated (post-validation) metrics.

## 5 Benchmark Design

### 5.1 Test Suite

The benchmark contains 115 test cases across five task groups (Table 2). Queries are written in natural and informal language: abbreviations (“*compare electricity '24 vs '25*”), colloquialisms (“*can ya tell me*”), misspellings (“*distribuition*”), and domain-specific terms (“*combinis*”, “*nomikais*”, “*shinkansen*”). This mimics the Japan-based personal finance context of the target application and tests robustness to spelling variations, colloquialisms, and cultural terms.

### 5.2 Complexity Scoring

Each test case is assigned a difficulty score:

$$C = n_p + d_{\text{date}} + d_{\text{cat}} + d_{\text{rel}} + d_{\text{multi}} + d_{\text{abbr}} + d_{\text{hol}} \quad (1)$$

where  $n_p$  = number of expected parameters;  $d_{\text{date}} \in \{0-4\}$  scores date specification complexity;  $d_{\text{cat}} \in \{0-2\}$  is the category normalisation distance;  $d_{\text{rel}} \in \{0, 1\}$  flags relative time expressions;  $d_{\text{multi}} \in \{0-2\}$  counts compound parameter groups;  $d_{\text{abbr}} \in \{0, 1\}$  flags abbreviated dates; and  $d_{\text{hol}} \in \{0, 2\}$  flags calendar knowledge requirements. Scores are binned into three tiers: **L1** ( $C \leq 4$ , 24 cases), **L2** ( $5 \leq C \leq 7$ , 45 cases), **L3** ( $C \geq 8$ , 46 cases).

### 5.3 Evaluation Metrics

We adopt the metric suite of Kavathekar et al. (2025):

- **FSP** (Function Selection Precision): Binary indicator of correct tool prediction.
- **ACS** (Argument Completeness Score):  $F_1$  on argument *name* overlap.
- **AVC** (Argument Value Correctness): Fraction of correctly named arguments whose values also match.
- **Task Accuracy**:  $F_1$  on the combined set  $\{\text{tool}\} \cup \{k=v\}$ , jointly capturing tool selection and parameter correctness.
- **Correct Ratio (CR)**: Strict binary indicator—1.0 only when Task Acc = 1.0, i.e. the entire function call is perfectly correct.

All metrics are averaged across 5 repetitions per test case (temperature = 0.1) with 95% bootstrap confidence intervals (Efron, 1979) ( $n = 2000$  resamples).

### 5.4 Error Taxonomy

Each prediction is classified into one of seven mutually exclusive error types: CORRECT, TOOL\_WRONG, PARSE\_FAILURE, PARAM\_MISSING, PARAM\_VALUE\_WRONG, PARAM\_EXTRA, FORMAT\_VIOLATION. This taxonomy helps diagnose specific failures over just aggregate accuracy, which is crucial for evaluating the safety profile of on-device financial assistants with respect to real-user harm.



| Model           | Family            | Params | RAM    |
|-----------------|-------------------|--------|--------|
| EXAONE-4.0-1.2B | EXAONE-4.0 (2025) | 1.2B   | 2.0 GB |
| Gemma-3-1B-it   | Gemma-3 (2025)    | 1B     | 1.6 GB |
| LFM2.5-1.2B     | LFM2.5 (2026)     | 1.2B   | 1.4 GB |
| MiniCPM5-1B     | MiniCPM (2024)    | 1B     | 1.5 GB |
| Qwen3.5-0.8B    | Qwen3.5 (2026)    | 0.8B   | 1.3 GB |
| Qwen3.5-2B      | Qwen3.5 (2026)    | 2B     | 2.1 GB |

Table 3: Models evaluated. All use Q8\_0 GGUF quantization via llama.cpp. RAM is average peak RSS.

## 6 Experimental Setup

### 6.1 Models

All six models (Table 3) use Q8\_0 GGUF quantization and run via llama.cpp (Gerganov, 2023) with full GPU offload on Apple Metal.

### 6.2 Hardware and Inference Protocol

Experiments run on an Apple M1 (8-core CPU, 8-core GPU, 8 GB unified memory). Parameters: temperature = 0.1, min\_p = 0.05, n\_ctx = 4096, stop tokens ["\n\n", "\n"]. Each model is warmed up through all six prompt templates (one router + five specialists) before evaluation begins. Each of the 115 test cases is run 5 times per model per architecture, yielding  $115 \times 5 \times 6 \times 2 = 6,900$  total observations.

## 7 Results

### 7.1 Overall Performance

Table 4 presents the full results. Figure 1 shows the accuracy-latency Pareto frontier and the error taxonomy breakdown.

**Key findings (RQ1, RQ2).** (1) Sub-2B SLMs can reliably call tools to perform personal finance analytics on consumer hardware (RQ1), with the Qwen3.5 family achieving >89% Task Accuracy in dual mode. This demonstrates that cloud reliance can be avoided while keeping all data-queries and results-private to the user’s device. (2) While single-agent prompts allow Qwen3.5-2B to reach 92.0% Task Accuracy, dual-agent decomposition is still preferred for two reasons (RQ2): (a) Strict correctness-Correct Ratio improves from 60.2% to 77.6% (+17.4 pp) for Qwen3.5-2B, representing a significant reduction in silent parameter errors that could produce incorrect outputs. (b) En-

abling smaller models-Qwen3.5-0.8B improves from 79.8% to 89.1% Task Accuracy (+9.3 pp) while almost halving latency (5.8 s to 3.0 s) in dual mode, making sub-1B models viable for low-memory devices.

### 7.2 Per-Tool-Group Analysis

Table 5 shows per-group accuracy for the top two dual-mode models. Figure 2 shows the full model  $\times$  category heatmap. `get_top_expenses` reaches 98.9% accuracy with Qwen3.5-2B due to its simple parameter structure. Conversely, `plot_time_series` is the most difficult task (74.4% for Qwen3.5-2B) due to complex date specifications and relative time expressions (e.g., “past X months”) requiring temporal reasoning.

### 7.3 Complexity Tier Analysis

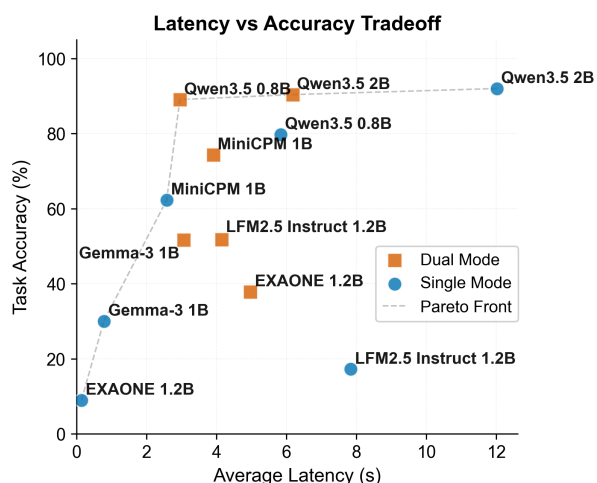
Figure 3 presents the per-model accuracy breakdown across complexity tiers in dual mode. Task Accuracy decreases from 96% (L1) to 82% (L3) for Qwen3.5-2B, and from 95% to 84% for Qwen3.5-0.8B. L3 queries involve constructs like cross-year range comparisons (“compare groceries nov 2024 to april 2025 vs nov 2025 to april 2026”), which require extracting up to 10 parameters with complex temporal constraints.

### 7.4 Error Taxonomy Analysis

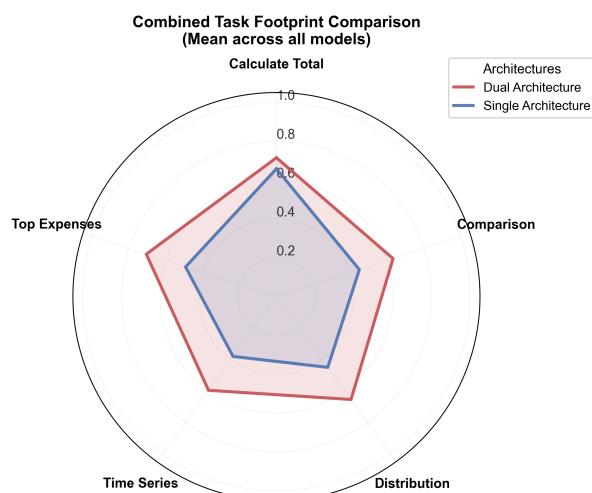
Figure 4 visualises the full error distribution. The error distribution shifts significantly between configurations (RQ3).

For Qwen3.5-2B: single-agent produces 22.6% PARAM\_EXTRA and 7.3% PARAM\_MISSING; dual-agent reduces PARAM\_EXTRA to 1.9% while PARAM\_MISSING remains at 8.0%.

The human-facing implications of this shift are significant. A PARAM\_EXTRA error is a silent corruption: when a model adds a spurious `months=1` filter to the query “how much did I spend on food?”, the system returns an answer- last month’s food spending- that looks correct but answers a different question. A user relying on this figure for tax preparation, expense reporting, or budgeting receives a plausible but wrong number, with no indication that the scope was silently restricted. In contrast, a PARAM\_MISSING error produces a visibly broader result (e.g., all-time food spending when a month was intended), which is more likely to trigger user scrutiny. The dual-agent architecture’s shift from extra-parameter to missing-parameter failures thus represents a qualitative improvement



(a) Accuracy-latency Pareto frontier. The dual-mode Qwen3.5-0.8B (orange square,  $\approx 3$  s, 89%) sits on the efficient frontier, offering the best accuracy per unit latency. Single-mode Qwen3.5-2B reaches the highest Task Accuracy (92%) but at  $4\times$  the latency (12 s).



(b) Task footprint (mean Task Accuracy across all models). The dual architecture (red) uniformly expands coverage over single (blue) on all five task groups, with the largest gains on time series and comparison tools.

Figure 1: System-level results. (a) Dual-agent Qwen3.5-0.8B dominates the efficiency frontier for privacy-preserving deployments. (b) Dual-agent decomposition improves coverage across all task categories.

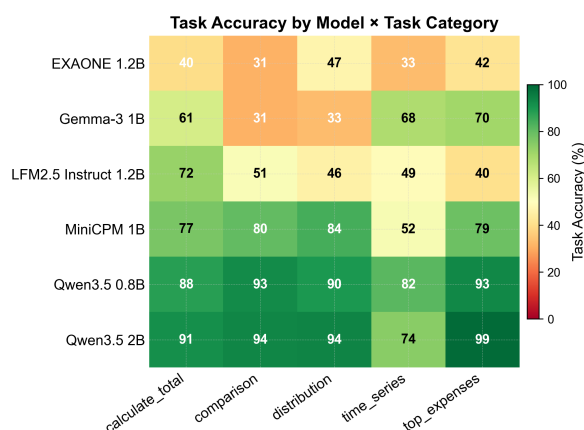


Figure 2: Task Accuracy (%) by Model  $\times$  Tool Category in dual mode. Qwen3.5 models achieve  $>74\%$  across all categories, while weaker models show marked gaps in comparison and time series tasks.

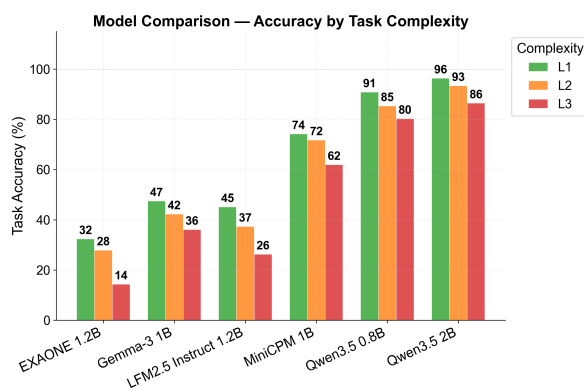


Figure 3: Task Accuracy by complexity tier (L1/L2/L3) per model in dual mode. The Qwen3.5 family degrades gracefully with increasing complexity.

in user-facing safety, not just a numerical accuracy improvement.

For Qwen3.5-0.8B: single-agent produces 32.5% PARAM\_MISSING and 11.8% PARAM\_EXTRA; dual-agent reduces these to 12.9% and 11.1% respectively, shifting the model from a high-variance to a more predictable failure profile.

## 7.5 Token Efficiency

Figure 5 shows that despite using  $\approx 2,700$  prompt tokens per query (vs.  $\approx 5,500$  for single-agent), the dual-agent architecture achieves equivalent or superior accuracy. The single-agent pipeline gets access

to larger contexts but exhibits lower performance per token since the model must simultaneously attend to all five tool schemas and disambiguation instructions. The dual-agent architecture decomposes the context, focusing each pass on a single subtask to achieve higher accuracy-per-token.

## 7.6 Validation Layer Impact

The validation layer (Table 6) contributes modest gains, primarily correcting category casing and normalisation. This reveals that argument extraction remains the primary bottleneck, since validation cannot recover missing parameters (PARAM\_MISSING).

| Mode   | Model             | FSP         | ACS  | AVC         | Task Acc    | CR          | Latency (s) | RAM (GB) |
|--------|-------------------|-------------|------|-------------|-------------|-------------|-------------|----------|
| Single | EXAONE-4.0-1.2B   | 19.3        | —    | 0.3         | 8.9         | 0.3         | 0.1         | 2.0      |
|        | Gemma-3-1B-it     | 27.0        | —    | 47.5        | 30.0        | 0.9         | 0.8         | 1.6      |
|        | LFM2.5-1.2B       | 28.7        | —    | 11.7        | 17.2        | 3.1         | 7.8         | 1.4      |
|        | MiniCPM5-1B       | 71.1        | —    | 86.7        | 62.3        | 13.0        | 2.6         | 1.5      |
|        | Qwen3.5-0.8B      | 95.0        | —    | 91.2        | 79.8        | 40.2        | 5.8         | 1.3      |
|        | <b>Qwen3.5-2B</b> | <b>96.5</b> | —    | <b>97.6</b> | <b>92.0</b> | 60.2        | 12.0        | 2.1      |
| Dual   | EXAONE-4.0-1.2B   | 90.6        | 3.7  | 3.3         | 37.8        | 3.0         | 5.0         | 1.9      |
|        | Gemma-3-1B-it     | 55.1        | 59.9 | 71.6        | 51.7        | 16.5        | 3.1         | 1.5      |
|        | LFM2.5-1.2B       | 69.6        | 51.6 | 57.2        | 51.8        | 16.3        | 4.1         | 1.4      |
|        | MiniCPM5-1B       | 84.9        | 77.0 | 81.6        | 74.3        | 43.5        | 3.9         | 1.5      |
|        | Qwen3.5-0.8B      | <b>94.3</b> | 90.2 | 96.8        | 89.1        | 61.7        | 3.0         | 1.2      |
|        | <b>Qwen3.5-2B</b> | 92.0        | 91.6 | <b>97.4</b> | 90.4        | <b>77.6</b> | 6.2         | 2.1      |

Table 4: Main results (validated metrics, %). **Bold** = best per metric. FSP = Function Selection Precision; ACS = Argument Completeness Score; AVC = Argument Value Correctness; Task Acc =  $F_1$ -based task accuracy; CR = strict exact-match Correct Ratio. ACS is omitted (—) for single-agent configurations where the joint output format makes isolated argument coverage less meaningful. Latency and RAM are per-query averages.

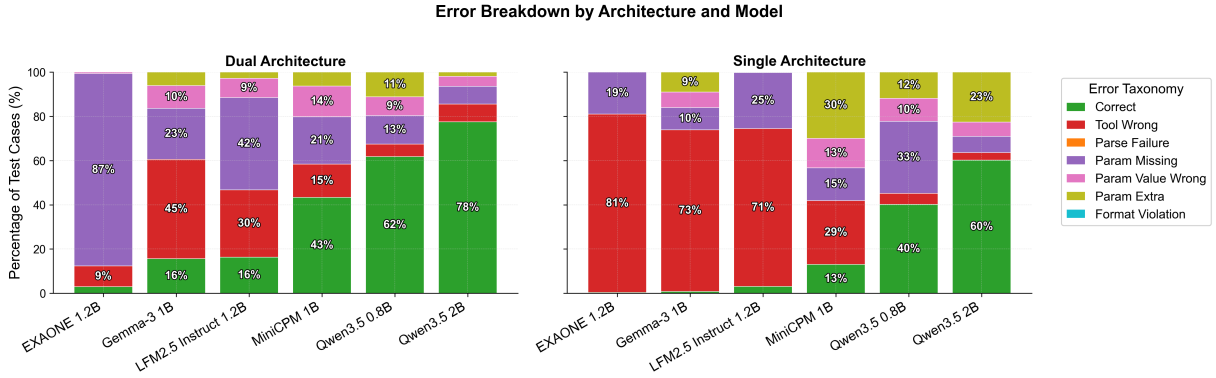


Figure 4: Error taxonomy breakdown (% of test cases) per model and architecture. Single-agent Qwen3.5-2B (right panel) produces 22.6% `PARAM_EXTRA` errors- spurious parameters that could lead to incorrect computations. Dual-agent (left) eliminates most of these, consolidating residual errors into the more predictable `PARAM_MISSING` category.

| Group        | Qwen3.5-0.8B |      | Qwen3.5-2B |      |
|--------------|--------------|------|------------|------|
|              | T.Acc        | CR   | T.Acc      | CR   |
| Time Series  | 81.6         | 54.8 | 74.4       | 60.9 |
| Distribution | 89.6         | 56.2 | 93.8       | 79.0 |
| Comparison   | 92.5         | 71.4 | 93.7       | 73.6 |
| Calc. Total  | 87.9         | 59.1 | 91.4       | 84.5 |
| Top Expenses | 93.3         | 64.8 | 98.9       | 92.4 |

Table 5: Per-group results in dual mode (validated, %). T.Acc = Task Accuracy, CR = Correct Ratio.

| Model (Dual)  | $\Delta$ Task Acc | $\Delta$ CR |
|---------------|-------------------|-------------|
| EXAONE-4.0    | +0.00             | +0.00       |
| Gemma-3-1B-it | +0.12             | +0.00       |
| LFM2.5-1.2B   | +0.50             | +0.00       |
| MiniCPM5-1B   | +1.17             | +0.87       |
| Qwen3.5-0.8B  | +0.41             | +1.57       |
| Qwen3.5-2B    | +0.29             | +0.87       |

Table 6: Validation layer impact ( $\Delta$  in percentage points, dual mode). Gains are modest but consistent for capable models.

## 8 Discussion

**Privacy through Task Decomposition.** The dual-agent architecture improves accuracy and makes local inference viable. By separating routing (single-digit output) from extraction (focused JSON), the dual-agent setup reduces the generation complexity per call, addressing the primary bottleneck of small models. Decomposing the task helps sub-2B models perform reliably on domain-specific schemas and by doing so, makes on-device inference a practical privacy guarantee rather than an aspirational one.

**Addressing Highly Complex Queries.** Even the best model only achieves 59.6% strict Correct Ratio on L3 queries in dual mode, which involves complex cross-year range comparisons requiring up to 10 interdependent parameters. Fine-tuning on

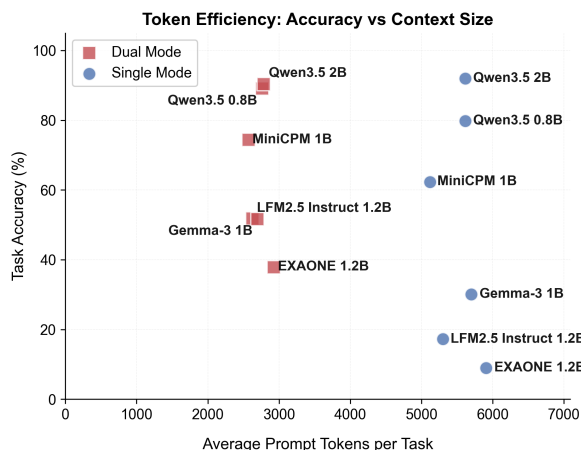


Figure 5: Token efficiency: Task Accuracy vs. average prompt tokens per query. Dual-agent configurations (red) achieve higher accuracy with fewer total tokens than their single-agent counterparts (blue), demonstrating superior context utilisation.

domain-specific examples using verified synthetic data (Liu et al., 2024b) could be a potential method to improve performance on complex queries.

**Deployment Recommendations.** Based on our evaluation, we recommend: (1) *Memory-constrained* ( $\leq 2$  GB available): Qwen3.5-0.8B dual, 89.1% Task Acc, 3.0 s, 1.2 GB RAM. (2) *Accuracy-first*: Qwen3.5-2B dual, 90.4% Task Acc, 77.6% CR, 6.2 s, 2.1 GB RAM. (3) *Budget-hardware*: MiniCPM5-1B dual, 74.3% Task Acc, 43.5% CR, 3.9 s, 1.5 GB RAM. All configurations keep user data entirely on-device.

**Case Study.** Figure 6 shows an authentic usage session of ExpenseSense. As a real-world deployment test in February 2026, an author used the system to query their own personal finance data in informal language—“can ya plot spending on snacks for past X months?”. The dual-agent Qwen3.5-0.8B pipeline correctly routed this to `plot_time_series` and extracted the parameters `{category: snacks, months: 14}` - all in 6 seconds. As shown in the plot, the resulting visualization revealed a clear trend in weekly snack expenditure. By surfacing this data locally, the author was able to reflect on their habits and successfully reduce their spending over time (from peaks of ¥3,000–4,000 to under ¥1,000). Crucially, this entire interaction- query, inference and plotting - remained strictly on the user’s device; no sensitive data was transmitted externally at any point. Connecting technical mechanisms to human needs, this demonstrates how private, local

analytics can directly support personal budgeting workflows.

## 9 Conclusion

We presented ExpenseSense, a human-centered evaluation of privacy-preserving, on-device financial tool calling using sub-2B quantized models. Our central finding is that local inference at this scale is not just technically feasible but practically reliable: Qwen3.5-0.8B in dual mode achieves 89.1% Task Accuracy with 3.0 s latency, keeping all user data—queries, context, and results—strictly on-device. Architectural decomposition into intent routing and parameter extraction further improves strict correctness (from 60.2% to 77.6% Correct Ratio) and, critically, shifts error modes toward safer failures: spurious parameter additions, which can silently corrupt financial analytics and mislead user decisions, are replaced by missing-parameter errors that are more visible and less harmful.

Several directions warrant future investigation. First, *fine-tuning* the best-performing models on domain-specific function-calling data could address the residual L3 failures while preserving on-device deployment. Second, extending the benchmark to *multi-turn* interactions would capture realistic conversational patterns, including follow-up queries where prior context affects parameter inference. Third, *user studies* examining how people understand and verify on-device privacy guarantees would complement our architectural analysis with evidence about user trust and mental models of local inference. Fourth, *multilingual* query support would broaden applicability across diverse user populations. Finally, evaluating lower-bit quantisation formats (Q4\_K\_M) could further reduce the memory footprint, making privacy-preserving financial AI accessible on lower capacity devices such as smartphones.

## Limitations

- All models use Q8\_0 quantisation; lower-bit formats (Q4\_K\_M) may yield different accuracy–efficiency tradeoffs.
- Test cases were authored by a single developer in English, reflecting a Japan-based context; crowdsourcing and multilingual support would improve diversity.
- Ground-truth labels are deterministic; some queries may admit legitimate alternative tool



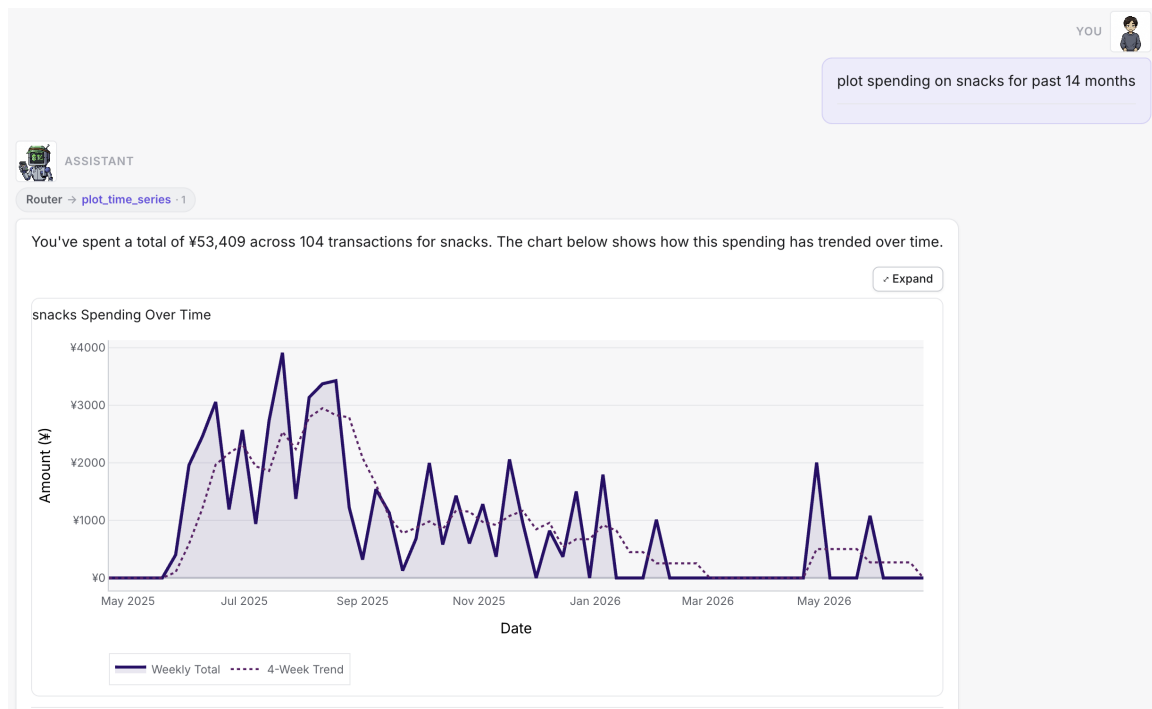


Figure 6: Real usage session: a user queries snack spending trends in informal language. The dual-agent Qwen3.5-0.8B pipeline correctly invokes `plot_time_series`, producing an actionable spending trend—entirely on-device with no data transmitted externally.

selections.

- We evaluate tool-call formation, not downstream financial calculation correctness.
- Our privacy guarantee is architectural (no data leaves the device) rather than cryptographic.

## Ethics Statement

ExpenseSense is designed for on-device inference specifically to preserve user financial privacy. No real user financial data was used in the benchmark; all test cases reference synthetic expense categories. The system is intended to complement, not replace professional financial advice. We recognise that even on-device models may produce incorrect financial summaries, and advocate for clear user-facing disclaimers in production deployments.

## Data and Code Availability

Code, prompts, benchmark data, and full results will be released on GitHub upon acceptance. Released artefacts include: all 115 test cases with ground-truth labels, the complete router and specialist prompt templates, inference configuration details, raw and validated CSV result files, and the benchmark runner, plotting scripts, and the exact model GGUFs used in this study to ensure complete reproducibility.

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## 710 A Complexity Scoring Features

| Feature            | Range | Description                                                 |
|--------------------|-------|-------------------------------------------------------------|
| $n_p$              | 0+    | Number of expected parameters                               |
| $d_{\text{date}}$  | 0–4   | Date complexity: none / year / month / range / specific day |
| $d_{\text{cat}}$   | 0–2   | Category distance: exact / alias / semantic                 |
| $d_{\text{rel}}$   | 0–1   | Relative time expression (“past X months”)                  |
| $d_{\text{multi}}$ | 0–2   | Compound parameter groups                                   |
| $d_{\text{abbr}}$  | 0–1   | Abbreviated year formats (“’24”)                            |
| $d_{\text{hol}}$   | 0–2   | Holiday/calendar knowledge required                         |

Table 7: Seven features of the complexity scoring function (Eq. 1).

## 711 B Sample Test Cases

| ID   | Query                                                          | Tool                 | Expected Params                                                                      | Tier |
|------|----------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------|------|
| TS01 | Can ya show spending on food for past 6 months?                | plot_time_series     | category=Food, months=6                                                              | L2   |
| DI06 | show breakdown of expenses for june 21 2024                    | plot_distribution    | year=2024, month=6, day=21                                                           | L2   |
| CP24 | compare groceries nov 2024 to apr 2025 vs nov 2025 to apr 2026 | plot_comparison_bars | category=grocery, y1=2024, sm1=11, ey1=2025, em1=4, y2=2025, sm2=11, ey2=2026, em2=4 | L3   |
| CT07 | sum total spent on metro for july 2024?                        | calculate_total      | category=tokyo metro, year=2024, month=7                                             | L2   |
| TP06 | top 5 expenses past month? disregard rent                      | get_top_expenses     | n=5, months=1, ignore_rent=true                                                      | L2   |

Table 8: Sample test cases. Note informal language, abbreviations, and semantic mappings (“metro”→“tokyo metro”, “groceries”→“grocery”).

## 712 C Qualitative Error Examples

713 Table 9 illustrates representative failures for the dominant error types in Qwen3.5-2B (dual mode).

| Error Type        | Query                                                            | Expected         | Model Output                                                                                                            |
|-------------------|------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------|
| PARAM_MISSING     | “Show spending trend for the last 3 months”                      | {months: 3}      | { } — months omitted in the relative-time case                                                                          |
| PARAM_VALUE_WRONG | “compare groceries nov 2024 to apr 2025 vs nov 2025 to apr 2026” | {em1: 4, em2: 4} | {em1: 11, em2: 11} — confuses start month for end month in cross-year range                                             |
| PARAM_EXTRA       | “How much did I spend on food?”                                  | {category: Food} | {category: Food, months: 1} — spurious time filter restricts scope to last month, silently returning an incorrect total |

Table 9: Representative failure cases for Qwen3.5-2B (dual mode). PARAM\_EXTRA is the most dangerous failure mode in financial contexts: the spurious months=1 filter silently restricts “total food spending” to “last month’s food spending”, returning a plausible but incorrect figure that a user may act on without realising the scope was changed.

714 **D Supplementary Figures**

715 Figure 7 provides the routing confusion matrix across tools, while Figure 8 shows the per-category  
716 footprint across all evaluated models. Figure 9 illustrates the metrics improvement provided by the  
717 post-hoc validation layer.

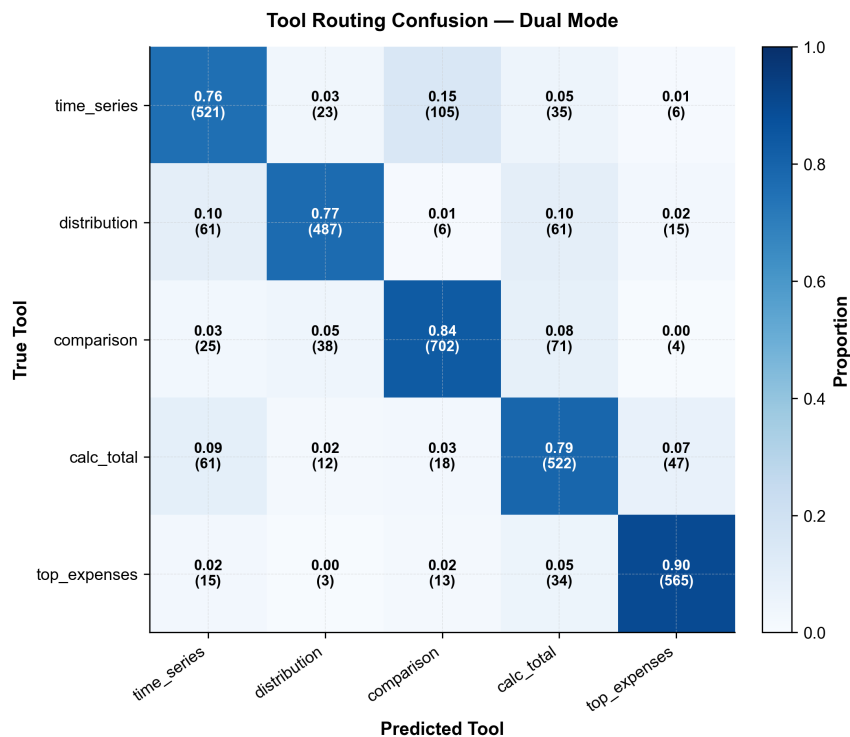


Figure 7: Normalized routing confusion matrix for the dual-agent router. The moderately high diagonal accuracy (76–90%) validates that the sequential classification step introduces negligible routing error propagation.



### Per-Model Task Footprint: Architecture Comparison

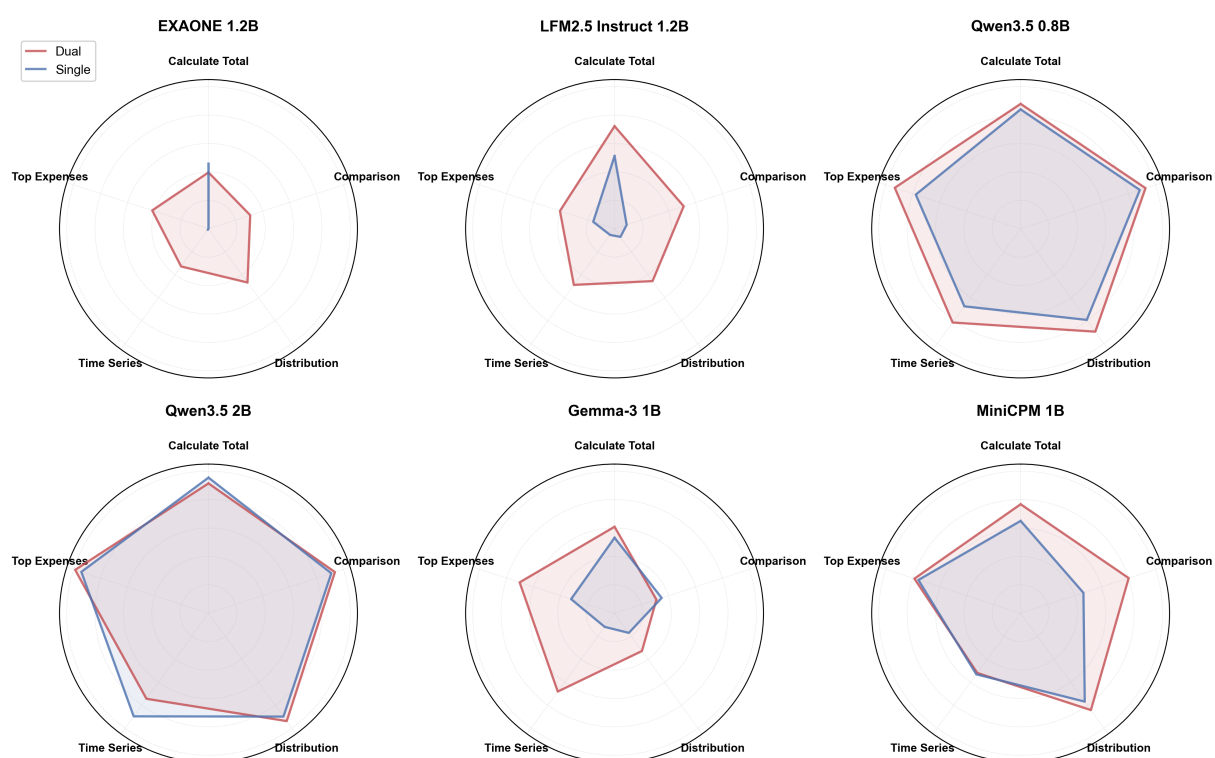


Figure 8: Per-model task footprint across all five task categories (Single vs. Dual mode). The dual-agent architecture provides near-universal improvement across models and tasks.

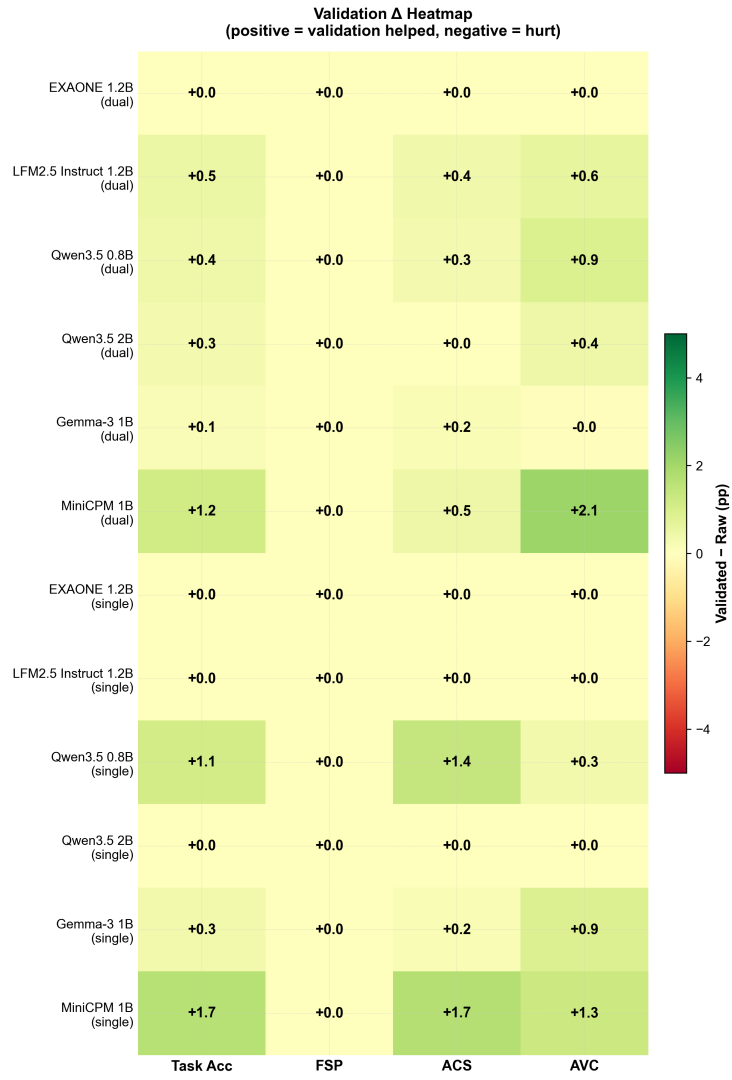


Figure 9: Validation  $\Delta$  Heatmap, showing absolute percentage point improvement across metrics after applying post-hoc fuzzy matching.

## 718 E Prompt Templates

719 The on-device prompts utilized for routing and parameter extraction tasks are detailed below.

### 720 E.1 Router Prompt

721 The router receives a focused intent-classification prompt:

```
722 You are an expert intent classifier for an expense analysis assistant.
723 Your job is to determine which ONE tool is best suited to answer the user's
724 question.
725
726 ## Available Tools
727
728 1. **`plot_time_series`** (ID: 1)
729 - Use when user asks about: trends, spending over time, "past X months",
730   "since 2023", date ranges.
731 - Keywords: trend, over time, plot, show spending, changed, since.
732 - NOTE: if user says "trend" or "over time" WITH "exclude rent", still
733   use this tool.
734
735 2. **`plot_distribution`** (ID: 2)
736 - Use when user asks for: breakdown, distribution, proportions, pie
737   chart, "how is X split", "what does X look like".
738 - Keywords: breakdown, distribution, distro, pie chart, split, look like.
739 - NOTE: if user says "breakdown" or "split" WITH "exclude rent", still
740   use this tool.
741
742 3. **`plot_comparisonBars`** (ID: 3)
743 - Use ONLY when comparing TWO different time periods (e.g. "Dec 2024 vs
744   Dec 2025" or ranges like "Jan-Jun 2024 vs Jan-Jun 2025").
745 - REQUIRES two distinct periods. If the user only mentions one month or
746   one year, DO NOT use this tool.
747 - Keywords: compare, contrast, vs, versus, difference between.
748
749 4. **`calculate_total`** (ID: 4)
750 - Use when user asks for: simple totals, sums, specific amounts "how
751   much did I spend".
752 - Keywords: how much, total, sum, cost.
753
754 5. **`get_top_expenses`** (ID: 5)
755 - Use when user asks for: biggest/largest expenses, top X items, most
756   expensive.
757 - Keywords: biggest, largest, top, most expensive, highest.
758 - NOTE: if user says "top X" WITH "exclude rent", still use this tool.
759
760 ## Output Format
761 Output ONLY the tool number (1-5). Do not output anything else, NO
762 MARKDOWN, NO BACKTICKS, NO EXPLANATION, NO CODE!
763
764 ## WRONG outputs (NEVER do this):
765 [X] I think the best tool is 4
766 [X] `4`
767 [X] Based on the query, I would use 1 because...
768 [X] ```4```
769 [X] The answer is: 2
770
771 ## CORRECT outputs:
772 [v] 4
773 [v] 1
774
775 ## Examples
776 User: "How much did I spend on food?"
777 Output: 4
778
779 User: "Total cost of internet for the last 3 months"
780 Output: 4
781
782 User: "How much did I spend in the last 6 months without rent?"
783 Output: 4
784
785 User: "Show me my spending trend for groceries"
786 Output: 1
787
788 User: "How has my rent changed over the last 6 months?"
789 Output: 1
790
791 User: "Show spending trend for past 4 months exclude rent tho"
792 Output: 1
793
794 User: "Compare food spending in 2024 and 2025"
795 Output: 3
796
797 User: "Difference in spending between Dec 2023 and Dec 2024"
798 Output: 3
799
800 User: "Compare overall spend in '24 vs '25 (exclude rent tho)"
```

```

801 Output: 3
802
803 User: "Compare dining between Jan-Jun 2024 and Jan-Jun 2025"
804 Output: 3
805
806 User: "Contrast spending from Nov 2024 to Feb 2025 vs Nov 2025 to Feb 2026"
807 Output: 3
808
809 User: "What were my biggest expenses?"
810 Output: 5
811
812 User: "Show me my top 5 largest purchases this year"
813 Output: 5
814
815 User: "Top 10 items without rent"
816 Output: 5
817
818 User: "Pie chart of transportation"
819 Output: 2
820
821 User: "How is my spending split in 2023?"
822 Output: 2
823
824 User: "Show me my spending breakdown excluding rent for the past 6 months"
825 Output: 2
826
827 User: "what does my electronics spending look like for jan 2025?"
828 Output: 2
829
830 User: "share distro of household expenses"
831 Output: 2

```

## 832 E.2 Specialist Base Template

833 Each specialist prompt instantiates a shared template defining output structure and relative date handling.  
834 The {metadata} placeholder is populated dynamically with the categories present in the user-uploaded  
835 dataset:

```

836 You are an automated API parameter extractor. Your ONLY purpose is to extract
837 parameters from the user's request and output a SINGLE JSON object containing
838 those parameters.
839
840 ## CRITICAL RULES (READ CAREFULLY)
841 1. DO NOT write any Python code or function calls. DO NOT import anything.
842 2. DO NOT wrap the output in markdown. NO backticks (`).
843 3. Output EXACTLY and ONLY a JSON object representing the parameters.
844    E.g. {"category": "Food", "months": 6}
845 4. Use EXACT category names from the metadata below. If not an exact match,
846    map it to the closest valid category (e.g. 'groceries' -> 'grocery',
847    'dining out' -> 'dining').
848
849 ## Date Rules
850 - 'months=N': RELATIVE duration (e.g. "past month" -> months=1, "last 3
851   months" -> months=3, "past year" -> months=12).
852 - Fractional years: convert to months. "past 0.5 years" -> months=6, "past
853   2.5 years" -> months=30.
854 - Abbreviated years: '24 = 2024, 24 = 2024, ALWAYS output the full 4-digit
855   year.
856 - 'year=YYYY': specific full calendar year.
857 - 'year=YYYY, month=M': specific calendar month (e.g. "Jan 2024").
858 - 'year=YYYY, month=M, day=D': specific calendar date.
859 - 'start_year=YYYY, end_year=YYYY': specific year range.
860 - 'start_year=YYYY, start_month=M, end_year=YYYY, end_month=M': specific
861   month-to-month range.
862 - For comparing two ranges (e.g., comparing Jan-Jun 2024 vs Jan-Jun 2025),
863   use comparison range parameters: 'y1=2024, sm1=1, em1=6, y2=2025, sm2=1,
864   em2=6'.
865 - For comparing cross-year ranges (e.g., Nov 2024-Jan 2025 vs Nov 2025-Jan
866   2026), use 'ey1' and 'ey2': 'y1=2024, sm1=11, ey1=2025, em1=1, y2=2025,
867   sm2=11, ey2=2026, em2=1'.
868 - CRITICAL: For relative queries like "past month", "last month", "last 6
869   months", ALWAYS use 'months=N'. Do NOT use 'month=M'.
870 - Pick ONLY ONE time filter. Do not mix 'months' with 'year'.
871
872 ## Available Parameters
873 {parameters}
874
875 ## Examples
876 {examples}
877
878 ## Context
879 ```
880 {metadata}
881 ```
882 Currency: JPY
883 Today: {current_date}
884

```

885 FINAL REMINDER: Output ONLY the JSON object. NO MARKDOWN, NO BACKTICKS,  
886 NO EXPLANATION, NO CODE!

### 887 E.3 Specialist Tool Prompts

888 Each tool-specific specialist appends its unique parameter definitions and few-shot examples to the base  
889 template. The precise specifications for each tool are detailed below.

#### 890 E.3.1 plot\_time\_series

```
891 Parameters:
892   category (str, optional), year (int, optional), month (int, optional),
893   start_year (int, optional), start_month (int, optional),
894   end_year (int, optional), end_month (int, optional),
895   months (int, optional), ignore_rent (bool, optional, defaults to false)
896
897 ## Examples
898 Q: "How much did I spend on futsal for the past 6 months?"
899 {"category": "futsal game", "months": 6}
900
901 Q: "Show me food spending from 2023 to 2025"
902 {"category": "Food", "start_year": 2023, "end_year": 2025}
903
904 Q: "Plot spending on snacks from oct 2024 to dec 2025"
905 {"category": "snacks", "start_year": 2024, "start_month": 10, "end_year": 2025, "end_month": 12}
906
907 Q: "Can ya plot spending on futsal on dec 2024?"
908 {"category": "futsal game", "year": 2024, "month": 12}
909
910 Q: "Gym expenses in 2024?"
911 {"category": "gym", "year": 2024}
912
913 Q: "Show me my spending trend for the last 3 months"
914 {"months": 3}
915
916 Q: "plot spending from june 2023 to dec 2025"
917 {"start_year": 2023, "start_month": 6, "end_year": 2025, "end_month": 12}
918
919 Q: "Trend of electricity bills since 2022"
920 {"category": "electricity bill", "start_year": 2022}
921
922 Q: "How has my transportation spending changed over time?"
923 {"category": "Transportation"}
924
925 Q: "Show spending on combinis for past year"
926 {"category": "combinini meal", "months": 12}
927
928 Q: "Show spending trend for past 4 months exclude rent tho"
929 {"months": 4, "ignore_rent": true}
930
931 Q: "plot spend on cafes for past 2.5 yrs"
932 {"category": "cafe", "months": 30}
933
934 Q: "show my ride share costs from '24 to 26"
935 {"category": "ride share", "start_year": 2024, "end_year": 2026}
936
937 Q: "show electronics spending since 2023"
938 {"category": "electronics", "start_year": 2023}
```

#### 939 E.3.2 plot\_distribution

```
940 Parameters:
941   category (str, optional), year (int, optional), month (int, optional),
942   day (int, optional), start_year (int, optional), start_month (int, optional),
943   end_year (int, optional), end_month (int, optional), months (int, optional),
944   ignore_rent (bool, optional, defaults to false)
945
946 ## Examples
947 Q: "Show me a breakdown of my food expenses in 2024"
948 {"category": "Food", "year": 2024}
949
950 Q: "Pie chart of all expenses from Nov 2024 to Feb 2025"
951 {"start_year": 2024, "start_month": 11, "end_year": 2025, "end_month": 2}
952
953 Q: "Show me my spending breakdown for last month"
954 {"months": 1}
955
956 Q: "Show me my spending breakdown for groceries for past 6 months"
957 {"category": "Grocery", "months": 6}
958
959 Q: "show spend distribution for 2026 feb"
960 {"year": 2026, "month": 2}
961
962 Q: "show breakdown of all expenses from 2024 to 2025"
963 {"start_year": 2024, "end_year": 2025}
964
```



```

965 Q: "Show me a breakdown of expenses for Dec 2024 (exclude rent tho)"
966 {"year": 2024, "month": 12, "ignore_rent": true}
967
968 Q: "Show spending breakdown for 2023/06/22"
969 {"year": 2023, "month": 6, "day": 22}
970
971 Q: "Spending distribution for the last 3 months"
972 {"months": 3}
973
974 Q: "Breakdown of transportation spending from 2022 to 2024"
975 {"category": "Transportation", "start_year": 2022, "end_year": 2024}
976
977 Q: "How did I split my money between categories in 2023?"
978 {"year": 2023}
979
980 Q: "Show me my spending breakdown excluding rent for the past 6 months"
981 {"months": 6, "ignore_rent": true}
982
983 Q: "Breakdown of fitness spending for 2025"
984 {"category": "Fitness", "year": 2025}
985
986 Q: "Breakdown of my spending since 2024"
987 {"start_year": 2024}
988
989 Q: "show breakdown of all expenses for mar 2026 (exclude rent tho)"
990 {"year": 2026, "month": 3, "ignore_rent": true}
991
992 Q: "what does my electronics and furniture spending look like for jan 2025?"
993 {"category": "Electronics and Furniture", "year": 2025, "month": 1}

```

### 994 E.3.3 plot\_comparison\_bars

```

995 Parameters:
996     category (str, optional), y1 (int, optional), m1 (int, optional),
997     d1 (int, optional), y2 (int, optional), m2 (int, optional),
998     d2 (int, optional), sm1 (int, optional), em1 (int, optional),
999     sm2 (int, optional), em2 (int, optional), ey1 (int, optional),
1000     ey2 (int, optional), ignore_rent (bool, optional, defaults to false)
1001
1002 ## Examples
1003 Q: "Compare food spending in 2024 vs 2025"
1004 {"category": "Food", "y1": 2024, "y2": 2025}
1005
1006 Q: "Compare dining Jan 2024 vs Jan 2025"
1007 {"category": "dining", "y1": 2024, "m1": 1, "y2": 2025, "m2": 1}
1008
1009 Q: "How does my spend on snacks compare between jan 2024 and jan 2026?"
1010 {"category": "snacks", "y1": 2024, "m1": 1, "y2": 2026, "m2": 1}
1011
1012 Q: "Compare total spending 2022 vs 2023"
1013 {"y1": 2022, "y2": 2023}
1014
1015 Q: "Compare overall expenses from jan 2025 vs jan 2026"
1016 {"y1": 2025, "m1": 1, "y2": 2026, "m2": 1}
1017
1018 Q: "Compare transportation on 21 July 2024 vs 21 July 2025"
1019 {"category": "Transportation", "y1": 2024, "m1": 7, "d1": 21, "y2": 2025, "m2": 7, "d2": 21}
1020
1021 Q: "Compare spend on electricity 2024 vs 2025"
1022 {"category": "electricity bill", "y1": 2024, "y2": 2025}
1023
1024 Q: "Compare gym spending Nov 2024 vs Nov 2025"
1025 {"category": "gym", "y1": 2024, "m1": 11, "y2": 2025, "m2": 11}
1026
1027 Q: "Compare my spending in 2024 vs 2025 excluding rent"
1028 {"y1": 2024, "y2": 2025, "ignore_rent": true}
1029
1030 Q: "compare education expenses '24 vs '25"
1031 {"category": "Education", "y1": 2024, "y2": 2025}
1032
1033 Q: "compare household expenses for april '24 vs apr 2025"
1034 {"category": "household", "y1": 2024, "m1": 4, "y2": 2025, "m2": 4}
1035
1036 Q: "Compare dining between Jan-Jun 2024 and Jan-Jun 2025"
1037 {"category": "dining", "y1": 2024, "sm1": 1, "em1": 6, "y2": 2025, "sm2": 1, "em2": 6}
1038
1039 Q: "compare electricity from jan-mar '24 vs jan-mar '25"
1040 {"category": "electricity bill", "y1": 2024, "sm1": 1, "em1": 3, "y2": 2025, "sm2": 1, "em2": 3}
1041
1042 Q: "Contrast overall spend from Nov 2024 to Feb 2025 vs Nov 2025 to Feb 2026"
1043 {"y1": 2024, "sm1": 11, "ey1": 2025, "em1": 2, "y2": 2025, "sm2": 11, "ey2": 2026, "em2": 2}
1044
1045 Q: "compare total spend on new years eve 2024 vs for same day on 2025"
1046 {"y1": 2024, "m1": 12, "d1": 31, "y2": 2025, "m2": 12, "d2": 31}

```

### 1047 E.3.4 calculate\_total

```

1048 Parameters:

```

```

1049     category (str, optional), year (int, optional), month (int, optional),
1050     day (int, optional), start_year (int, optional), start_month (int, optional),
1051     end_year (int, optional), end_month (int, optional), months (int, optional),
1052     ignore_rent (bool, optional, defaults to false)
1053
1054 ## Examples
1055 Q: "How much did I spend on groceries in Dec 2024?"
1056 {"category": "grocery", "year": 2024, "month": 12}
1057
1058 Q: "Total spending from Oct 2024 to March 2025"
1059 {"start_year": 2024, "start_month": 10, "end_year": 2025, "end_month": 3}
1060
1061 Q: "What is my total spending in 2025?"
1062 {"year": 2025}
1063
1064 Q: "How much did I spend on food in past month?"
1065 {"category": "Food", "months": 1}
1066
1067 Q: "Total cost of electricity in 2023"
1068 {"category": "electricity bill", "year": 2023}
1069
1070 Q: "Total spent on 15 Jan 2024?"
1071 {"year": 2024, "month": 1, "day": 15}
1072
1073 Q: "How much spent on rent in the last 6 months?"
1074 {"category": "rent", "months": 6}
1075
1076 Q: "Sum of all transportation expenses in 2025"
1077 {"category": "Transportation", "year": 2025}
1078
1079 Q: "Total spent on combini food for 2025?"
1080 {"category": "combini meal", "year": 2025}
1081
1082 Q: "How much did I spend in the last 6 months without rent?"
1083 {"months": 6, "ignore_rent": true}
1084
1085 Q: "Total spent on dining since 2024"
1086 {"category": "dining", "start_year": 2024}
1087
1088 Q: "whats the total spend on donations from '23 to 2026?"
1089 {"category": "donation", "start_year": 2023, "end_year": 2026}
1090
1091 Q: "Excluding rent, what was my total spend in 2025?"
1092 {"year": 2025, "ignore_rent": true}

```

### 1093 E.3.5 get\_top\_expenses

```

1094 Parameters:
1095     n (int, optional, defaults to 10), category (str, optional),
1096     year (int, optional), month (int, optional), day (int, optional),
1097     start_year (int, optional), start_month (int, optional),
1098     end_year (int, optional), end_month (int, optional),
1099     months (int, optional), min_amount (int, optional),
1100     ignore_rent (bool, optional, defaults to false)
1101
1102 ## Examples
1103 Q: "What were my biggest expenses in Dec 2024?"
1104 {"n": 10, "year": 2024, "month": 12}
1105
1106 Q: "Top 5 food expenses in 2024?"
1107 {"n": 5, "category": "Food", "year": 2024}
1108
1109 Q: "Top 10 food expenses in june 2025?"
1110 {"n": 10, "category": "Food", "year": 2025, "month": 6}
1111
1112 Q: "Top 5 expenses for 26 june 2024?"
1113 {"n": 5, "year": 2024, "month": 6, "day": 26}
1114
1115 Q: "Top expenses from July 2023 to Dec 2024"
1116 {"start_year": 2023, "start_month": 7, "end_year": 2024, "end_month": 12}
1117
1118 Q: "Show my top 10 expenses from 2023 to 2025"
1119 {"n": 10, "start_year": 2023, "end_year": 2025}
1120
1121 Q: "Top 10 expenses of past month?"
1122 {"n": 10, "months": 1}
1123
1124 Q: "What are my top expenses excluding rent?"
1125 {"n": 10, "ignore_rent": true}
1126
1127 Q: "Show my top 3 largest transactions in 2023"
1128 {"n": 3, "year": 2023}
1129
1130 Q: "What were the biggest expenses over the last 3 months, without rent?"
1131 {"n": 10, "months": 3, "ignore_rent": true}
1132
1133 Q: "Top 7 futsal expenses for 2025?"

```

1134 {"n": 7, "category": "futsal game", "year": 2025}  
1135  
1136 Q: "Top 5 expenses last month, exclude rent"  
1137 {"n": 5, "months": 1, "ignore\_rent": true}  
1138  
1139 Q: "What were my biggest expenses since 2023?"  
1140 {"n": 10, "start\_year": 2023}  
1141  
1142 Q: "show me top 10 expenses from jan 2025 to june '25"  
1143 {"n": 10, "start\_year": 2025, "start\_month": 1, "end\_year": 2025, "end\_month": 6}  
1144  
1145 Q: "tell me the top 15 expenses for 2025, ignore rent plz"  
1146 {"n": 15, "year": 2025, "ignore\_rent": true}